



next scarce target

arms of equal total size, same seed set

A. No expansion
the seed photos
as they are

**B. Classical offline
augmentation**
flip, rotation, scale-shift,
HSV; geometry and colour only.
Labels follow the transform, so
there is no review step

**C. State-editing
synthesis**
an image-editing model
rewrites the hazard state
of the same scenes

Auto-labelling + human review
quality gate: a labeler model
pre-labels every generated
image and a human inspects
and corrects each one

Unified training protocol
six detector families, identical hyperparameters,
fixed schedule with periodic snapshots

evaluated once, reported three ways

Independent-source development benchmark — quality gate
no source or scene overlap with the seed set, supported by a
reproducible thumbnail-similarity check

Three separate reports
fixed endpoint | cross-fitted held-out
predictions (pooled OOF) | deployment model

Deployment and reuse
scarce-hazard detection on site; the recipe is
re-instantiated for another scarce target